But the purpose of this section is to focus on building complex accessor functions to create, from scratch, the kinds of data visualization you've seen and likely thought of as exotic. Let's assume this data is correct and take a moment to analyze the effectiveness of this admittedly attractive method of visualizing data. Is this really a better way to show movie grosses than a simpler stacked graph or line chart? That depends on the scale of the questions being addressed by the chart. If you're trying to discover overall patterns of variation in movie grosses, as well as spot interactions between them (for instance, seeing if a particularly high-grossing-over-time movie interferes with the opening of another movie), then it may be useful. If you're trying to impress an audience with a complex-looking chart, it would also be useful. Otherwise, you'll be better off with something simpler than this. But even if you only build less-visually impressive charts, you'll still use the same techniques we've gone over in this section.

4.6 Summary

In this chapter you've learned the basics of creating charts:

- Integrating generators and components with the selection and binding process
- Learning about D3 components and the axis component to create chart elements like an x-axis and a y-axis
- Interpolating graphical elements, such as lines or areas from point data, using D3 generators
- Creating complex SVG objects that use the `<g>` element's ability to create child shapes, which can be drawn based on the bound dataset, using `.each()`
- Exploring the representation of multidimensional data using boxplots
- Combining and extending these methods to implement a sophisticated charting method, the streamgraph, while learning how such charts may outstrip their audience's ability to successfully interpret such data

These skills and methods will help you to better understand the D3 layouts, which we'll explore in more detail in the following chapters. The incredible breadth of data visualization techniques possible with D3 is based on the fundamental similarity between different methods of displaying data, at the visual level, at the functional level, and at the data level. By understanding how the processes work and how they can be combined to create more interactive and rich representation, you'll be better equipped to choose and deploy the right one for your data.